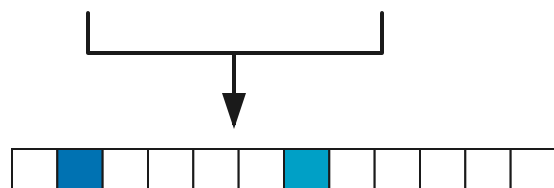


1 Measurement

held-out variant,
single cells



wild-type cells

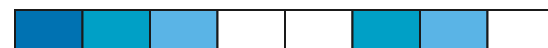


δ_v = pseudobulk target (ground truth)

mean(variant) - mean(WT)

2 Model

variant features (θ / ESM)



not the variant
identity, not
its cells

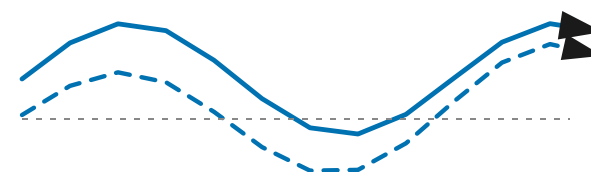
feature $\rightarrow$
response model



δ_v (predicted profile)

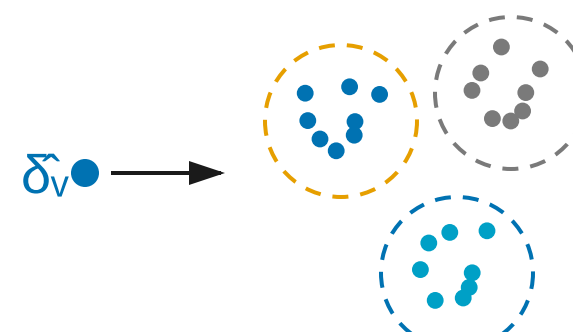
3 Evaluation

Direction (Pearson- δ)



points the right way

Discrimination (PDS)



nearest its own allele among siblings

5 Verdict

Verdict

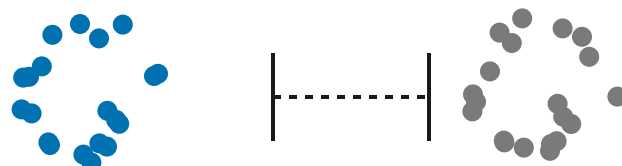
- ✓ measurable?
- ✓ benchmarkable?
- ✓ worth modelling?

expand redesign exclude

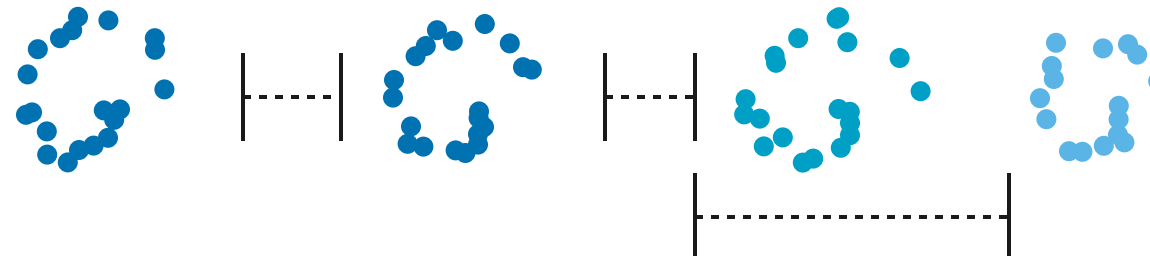
4 Resolution diagnostic (single cell)

detection = variant vs wild type

same single cells,
no averaging



identification = variant vs siblings



can the ground
truth reward an
allele model?